`cqrfdr`: FDR-controlled variable selection for high-dimensional censored quantile regression

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1 The problem

We observe $Y_i = \min(\log T_i, \log C_i)$, $\delta_i = 1\{T_i \le C_i\}$ and covariates $X_i \in \mathbb{R}^p$, with p possibly larger than n, and assume the linear quantile model $Q_{\log T}(u \mid X) = \beta_0(u) + X^\top \beta(u)$ for $u \in [\nu, \tau]$. The goal is to select the covariates with $\beta_j(\tau) \neq 0$ while controlling the false discovery rate $\mathrm{FDR} = E[\#\text{false selections}/(\#\text{selections} \vee 1)]$ at a nominal level q.

`ssms()` implements the single-split mirror-statistic procedure (SS-MS):

1. Split the data at random into halves D_1 and D_2.
2. On D_1, fit the L_1-penalized sequential Peng–Huang estimator (L-HDCQR) on the grid $\nu = \tau_0 < \cdots < \tau_L = \tau$, with the penalty multiplier chosen by cross-validation, and form a screened set of at most $\lfloor d/\log d \rfloor$ candidates (d = number of events).
3. On D_2, refit the screened model without penalty.
4. Combine the two estimates at τ into mirror statistics $M_j = \mathrm{sign}(\hat{b}_j^{(1)}\hat{b}_j^{(2)})(|\hat{b}_j^{(1)}| + |\hat{b}_j^{(2)}|)$ and select $\{j : M_j \ge \hat{t}\}$, where $\hat{t}$ is the smallest t with $(1 + \#\{M_j \le -t\})/(\#\{M_j \ge t\} \vee 1) \le q$.

2 A simulated example

`sim_cqr()` generates data from the designs of the paper's simulation study. Here: $n = 300$, $p = 200$, ten signals of size 0.8 with alternating signs, AR(1) covariates with $\rho = 0.5$, and 30% censoring.

```
> library(cqrfdr)
> set.seed(2026)
> d <- sim_cqr(n = 300, p = 200, s = 10, beta = 0.8, rho = 0.5, cens = 0.3,
+              calib_n = 50000)
> c(censoring = d$cens_rate, events = sum(d$delta))
```

```
censoring    events
     0.28    216.00
```

```
> d$beta[2:13]
```

```
  X1   X2   X3   X4   X5   X6   X7   X8   X9  X10  X11  X12
 0.8 -0.8  0.8 -0.8  0.8 -0.8  0.8 -0.8  0.8 -0.8  0.0  0.0
```

One call gives the selection at both FDR levels (only the threshold depends on q). The grid has 20 points here to keep the vignette fast; the default is 40.

```
> set.seed(1)
> fit <- ssms(d$x, d$y, d$delta, tau = 0.5, q = c(0.1, 0.2), n_grid = 20)
> fit
```

```
SS-MS: single split, mirror statistics
  censored quantile regression at tau = 0.5; n = 300 (216 events), p = 200
  grid of 20 quantile levels on [0.1, 0.5]
  screen: 23 candidates (rule "path", cap 23); penalty multiplier 0.2

   q selected threshold
 0.1       10    1.1960
 0.2       11    0.1793

Selected at q = 0.1:
  X1 X2 X3 X4 X5 X6 X7 X8 X9 X10

Selected at q = 0.2:
  X1 X2 X3 X4 X5 X6 X7 X8 X9 X10 X133
```

```
> truth <- paste0("X", d$active)
> rbind(`q = 0.1` = fdp_power(selected(fit, q = 0.1), truth),
+       `q = 0.2` = fdp_power(selected(fit, q = 0.2), truth))
```

```
        selected false true     FDP power
q = 0.1       10     0   10 0.00000     1
q = 0.2       11     1   10 0.09091     1
```

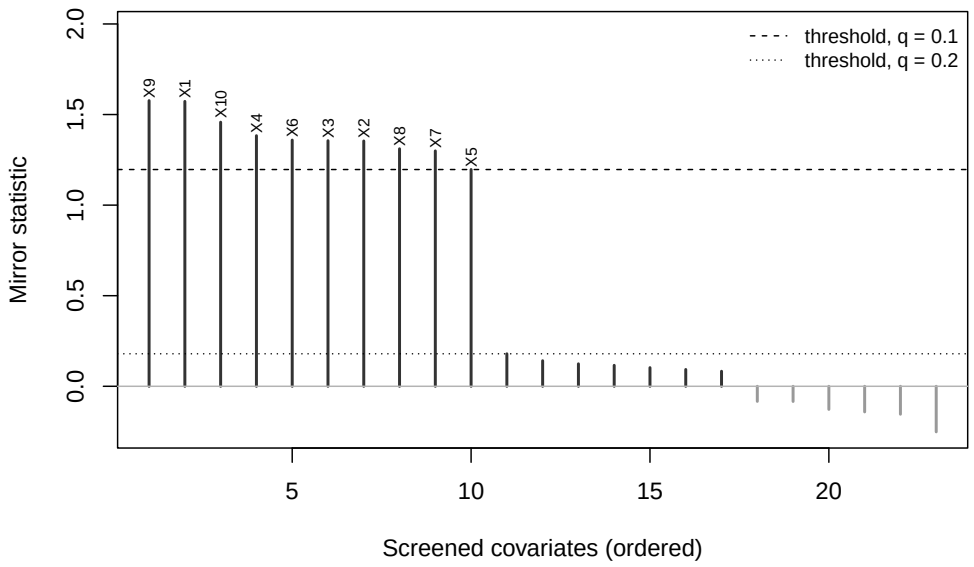
`summary()` lists the screened candidates with both half-sample coefficients and the mirror statistic; `plot()` shows the ordered mirror statistics with the two thresholds. Null covariates have mirror statistics symmetric about zero, which is what the threshold exploits.

```
> head(summary(fit), 12)
```

```
   variable index       b1       b2      M sel_q0.1 sel_q0.2
1        X9     9  0.55366  1.02318 1.5768     TRUE     TRUE
2        X1     1  0.73737  0.83521 1.5726     TRUE     TRUE
3       X10    10 -0.60667 -0.85118 1.4579     TRUE     TRUE
```

```
4         X4     4 -0.59946 -0.78378 1.3832     TRUE     TRUE
5         X6     6 -0.57193 -0.78708 1.3590     TRUE     TRUE
6         X3     3  0.47450  0.88114 1.3556     TRUE     TRUE
7         X2     2 -0.51602 -0.83815 1.3542     TRUE     TRUE
8         X8     8 -0.43474 -0.87627 1.3110     TRUE     TRUE
9         X7     7  0.54131  0.75794 1.2993     TRUE     TRUE
10        X5     5  0.50188  0.69458 1.1965     TRUE     TRUE
11      X133   133  0.12868  0.05061 0.1793    FALSE     TRUE
12       X86    86  0.07404  0.06706 0.1411    FALSE    FALSE
```

```
> plot(fit)
```



3 Main options

`tau`, `nu`, `n_grid` target quantile and the grid `seq(nu, tau, length.out = n_grid)`. The target must be estimable under the censoring: if few subjects remain at risk at the τ-th quantile, the path stops with status `"stage1_no_root"`.

`lambda` `"cv"` (default): K-fold CV of the held-out Peng–Huang loss over multipliers of the pivotal penalty; `"pivotal"`: a fixed multiplier (`fixed_mult`), much faster.

`screen` `"path"` (default) fills the candidate budget in the order in which covariates enter the lasso path below the CV penalty, which protects the screening of moderate signals; `"cv"` keeps the CV support only.

`screen_cap` `"d2"` (default, the paper) or `"d1"` (events in the first half, so the screen depends on D_1 only), an integer, or `"none"`.

`offset` 1 for the threshold U_+ with the finite-sample bound; 0 for the threshold of Dai et al. (2023). With `offset = 1` a nonempty selection has at least $\lceil 1/q \rceil$ elements, so with fewer than $1/q$ signals SS-MS selects nothing.

`cores` parallel CV fits on Unix-alikes.

Results depend on the random split, the folds and the Monte Carlo pivotal penalty; set a seed for reproducibility.

4 Multiple splitting: MS-MS

The selection of SS-MS depends on the random split. `msms()` runs SS-MS on B independent random splits and aggregates the selections through the inclusion rates of Dai et al. (2023), $I_j = B^{-1}\sum_b 1\{j \in \hat{S}^{(b)}\}/(|\hat{S}^{(b)}| \vee 1)$: the covariates with the smallest rates are removed until their total rate exceeds q. The penalty multiplier is chosen once by CV and reused in all splits, and ties are broken at random. The paper uses $B = 50$; $B = 10$ keeps the vignette short.

```
> set.seed(2)
> fs <- msms(d$x, d$y, d$delta, tau = 0.5, q = c(0.1, 0.2), B = 10, n_grid = 20)
> fs

MS-MS: SS-MS on B = 10 splits, aggregated by inclusion rates
  censored quantile regression at tau = 0.5; n = 300 (216 events), p = 200
  grid of 20 quantile levels on [0.1, 0.5]
  penalty multiplier 0.2 (cv_once); 0 of 10 splits failed

   q selected empty_splits
 0.1       10           0%
 0.2        9           0%

Selected at q = 0.1:
  X1 X2 X3 X4 X5 X6 X7 X8 X9 X10

Selected at q = 0.2:
  X1 X2 X3 X4 X6 X7 X8 X9 X10

> rbind(`q = 0.1` = fdp_power(selected(fs, 0.1), truth),
+       `q = 0.2` = fdp_power(selected(fs, 0.2), truth))

        selected false true FDP power
q = 0.1       10     0   10   0   1.0
q = 0.2        9     0    9   0   0.9

> fs$share_empty      # share of splits whose own SS-MS selection was empty

0.1 0.2
  0   0
```

MS-MS is more conservative than SS-MS: covariates selected in only a few splits are dropped. Two diagnostics are useful on real data. A large share of empty splits means that the cutoff U_+ is often infeasible (too few signals, or a screening cap that is too small; see below). Running `msms()` twice with different seeds and comparing the selections (for example by the Jaccard index) shows how much of the selection is due to the random splits.

5 Comparators

The comparators of the paper are available with the settings of their original publications.

- `ds()`: the single data-splitting procedure of Dai et al. (2023): candidate set = support of the cross-validated lasso (10-fold CV, no cap), unpenalized refit, cutoff without the offset.
- `mds()`: their multiple data-splitting procedure: B complete DS fits, each with its own 10-fold CV, aggregated with their selection rule.
- `hdcqr(lambda = "zheng")`: L-HDCQR with the penalty sequence of Zheng, Peng and He (2018), whose support at τ is a selection without FDR guarantee. `hdcqr()` with the default `lambda = "cv"` uses one cross-validated penalty level instead and also returns coefficients and predictions.

```
> set.seed(3)
> fd <- ds(d$x, d$y, d$delta, tau = 0.5, q = c(0.1, 0.2), n_grid = 20)
> fdp_power(selected(fd, 0.2), truth)

selected    false     true      FDP    power
 15.0000   5.0000  10.0000   0.3333   1.0000

> fz <- hdcqr(d$x, d$y, d$delta, tau = 0.5, n_grid = 20, lambda = "zheng")
> fdp_power(selected(fz), truth)

selected    false     true      FDP    power
 14.0000   4.0000  10.0000   0.2857   1.0000

> fh <- hdcqr(d$x, d$y, d$delta, tau = 0.5, n_grid = 20)
> round(coef(fh, tau = c(0.25, 0.5))[1:6, ], 3)

            0.2474 0.5000
(Intercept)  0.215  0.481
X1           0.704  0.675
X2          -0.625 -0.586
X3           0.721  0.729
X4          -0.717 -0.744
X5           0.670  0.636

> ## each of the B splits runs its own 10-fold CV, so this takes several minutes
> fm <- mds(d$x, d$y, d$delta, tau = 0.5, q = c(0.1, 0.2), B = 50)
```

6 Using your own data

The response can be a `survival::Surv` object, in which case the times are log-transformed. Covariates with no variation on a half sample must be removed first; heavy-tailed covariates can be standardized by median and MAD with `std = "robust"`.

```
> library(survival)
> X <- as.matrix(my_data[, candidate_columns])
> fit <- msms(X, Surv(my_data$time, my_data$status), tau = 0.5, q = 0.1, cores = 4)
> selected(fit, q = 0.1)
```

Two conditions behind the FDR guarantee can be checked before any selection is examined. First, the cutoff U_+ needs at least $1/q$ selections; with d_2 events in the second half, the candidate set has at most $\lfloor d_2 / \log d_2 \rfloor$ covariates, and the paper requires this cap to be at least 25, so that 10 selections at $q = 0.1$ do not need most of the candidates to be signals. Second, the first-stage screen should retain the active covariates. A plasmode simulation on the observed covariate matrix checks both: draw an active set, generate responses from the linear quantile model with censoring calibrated to the data, and record the FDR and power.

```
> plasmode_rep <- function(X, cens_rate, s = 10, b = 0.1, tau = 0.5, q = 0.1) {
+   Z <- scale(X); n <- nrow(Z)
+   act <- sort(sample(ncol(Z), s))
+   logT <- drop(Z[, act] %*% (b * sample(c(-1, 1), s, TRUE))) + rnorm(n)
+   g <- rnorm(n)
+   c0 <- uniroot(function(c) mean(c + g < logT) - cens_rate, range(logT) + c(-10, 10))$root
+   y <- pmin(logT, c0 + g); delta <- as.integer(logT <= c0 + g)
+   f <- msms(Z, y, delta, tau = tau, q = q)
+   fdp_power(selected(f, q), colnames(Z)[act])
+ }
```

7 A small Monte Carlo study

The code below estimates the FDR and power of SS-MS over replicated data sets, reusing the calibrated censoring location `c0`.

```
> reps <- 100
> res <- t(vapply(seq_len(reps), function(r) {
+   set.seed(r)
+   dr <- sim_cqr(n = 300, p = 200, beta = 0.8, cens = 0.3, c0 = d$c0)
+   f <- ssms(dr$x, dr$y, dr$delta, q = 0.2)
+   fdp_power(selected(f, 0.2), paste0("X", dr$active))
+ }, numeric(5)))
> colMeans(res[, c("FDP", "power")])   # FDR and average power
```

References

Dai, C., Lin, B., Xing, X. and Liu, J. S. (2023). False discovery rate control via data splitting. *Journal of the American Statistical Association*.

Peng, L. and Huang, Y. (2008). Survival analysis with quantile regression models. *Journal of the American Statistical Association*, 103, 637–649.

Zheng, Q., Peng, L. and He, X. (2018). High dimensional censored quantile regression. *The Annals of Statistics*, 46, 308–343.